MIL-C-52464B(ME)
26 July 1982
SUPERSEDING
MIL-C-52464A(ME)
4 November 1974

MILITARY SPECIFICATION

CYLINDER ASSEMBLIES, ACTUATING, LINEAR: HYDRAULIC, FOR BRIDGE AND BRIDGE LAUNCHER

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hydraulic cylinder assemblies for the M48A2, M48A5, and M60A1 Bridge Launcher-Transporters and the hydraulic cylinder assembly for the Armored Vehicle Launched Bridge.

1.2 Classification. The cylinder assemblies shall be of the following types and styles, as specified (see 6.2):

Type I - Holddown, M48A2/M48A5/M60A1.
Type II - Swing, M48A2
Type III - Scissoring, Bridge Only.
Type IV - Tongue
  Style A - M48A2
  Style B - M48A5/M60A1
Type V - Overhead
  Style A - M48A2
  Style B - M48A5/M60A1
Type VI - Ejection, M48A5/M60A1
Type VII - Locking, M48A2/M48A5/M60A1
Type VIII - Release M48A2

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the specifications and standrds of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	- Pallets, Material Handling, Wood, Stringer Construction 2 Way and 4 Way Partial.
QQ-S-781	- Strapping, Steel and Seals.
PPP-B-621	- Box, Wood, Nailed and Lock Corner.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment of Painting of Material.
MIL-H-46170	- Hydraulic Fluid, Rust Inhibited Fire-Resistant, Synthetic Hydrocarbon Base.
MIL-C-52950	- Crates, Wood, Open and Covered.

STANDARDS

FEDERAL

FED-STD-101	- Preservation, Packaging, and Packing Materials: Test Procedures.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

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DRAWINGS

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TA13211E3020	- Cylinder, Hold-Down.
TA13211E3130	- Cylinder, Locking.
TA13211E3140	- Cylinder, Ejection.
TA13211E7880	- Cylinder, Scissoring, Bridge.
TA13214E0000	- Cylinder, Swing, M48A2.
TA13214E0010	- Cylinder, Overhead, M48A2.
TA13214E0030	- Cylinder, Release, M48A2.
TA13214E0070	- Cylinder, Overhead.
TA13214E0080	- Cylinder, Tongue.
TA13214E0090	- Cylinder, Tongue, M48A2.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specifications to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc, 2501 N.W. Seventh Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Welding Qualifications, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The cylinder assemblies or separately procured cylinder components shall be in accordance with the applicable top assembly or drawings and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is

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permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the cylinders or separately procured cylinder components. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (Preproduction Model). The contractor shall furnish one or more of each type or each type and style cylinder assembly or one or more of each separately procured component as specified (see 6.2), for examination testing within the time frame specified (see 6.2), to prove, prior to starting production that his production methods will produce bridge launchers or separately procured components that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and unless otherwise specified herein, all examination and tests shall be conducted by the contractor and shall be subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Materials. Materials shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Copies of analyses and test reports for all materials shall be available for inspection. Chemical composition and mechanical properties of heat-treated materials shall be verified by chemical analysis and mechanical tests (see 6.9).

3.4 Cylinders. Each cylinder assembly shall operate at a 3,800 psi sustained working pressure without malfunctioning or leakage. Each cylinder assembly shall withstand the applicable proof pressure specified in table I, without external leakage or visible damage. The rate of internal leakage past the piston of each cylinder assembly at 3,800 psi pressure shall be not more than that specified in table I. All cylinder assemblies shall be filled with hydraulic oil conforming to MIL-H-46170. All hydraulic cylinder assemblies having the same classification (see 1.2), shall be identical.

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TABLE I. Hydraulic Cylinder Requirements.

Cylinder	Top Assembly	Application				Rate of Internal Leakage fl-oz/min	Proof Pressure psi
		M48A2	M48A5	M60A1	Bridge		
Type I	TA13211E3020	X	X	X		4	7,000
Type II	TA13214E0000	X				10	7,000
Type III	TA13211E7880				X	10	7,000
Type IV, Style A	TA13214E0090	X				12	7,000
Type IV, Style B	TA13214E0080		X	X		13	7,000
Type V, Style A	TA13214E0010	X				10	7,000
Type V, Style B	TA13214E0070		X	X		13	7,000
Type VI,	TA13214E3140		X	v		4	7,000
Type VII	TA13211E3130	X	X	X		4	7,000
Type VIII	TA13214E0030	X				N/A	5,000

3.5 Identification marking. The cylinders shall be identified in accordance with MIL-STD-130 and the markings indicated on the drawings.

3.6 Stenciling. The nomenclature, National Stock Number, weight, and cubage shall be stenciled on each cylinder assembly. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. Markings shall be Gothic-type capitals and Arabic numerals 3/4 inch high. Stenciling shall conform to MIL-T-704.

3.7 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.5):

Item No.	Quantity	Description	Identification
1	1 set	Inspection Equipment for Cylinders.	Quality Assurance Pamphlet, P715-124 This pamphlet is included in the Top Assembly.

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3.8 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.9 Type I holddown cylinder for M48A2/M48A5/M60A1 launchers. The type I cylinder assembly shall be in accordance with Top Assembly, TA13211E3020, and as specified herein.

3.10 Type II, swing cylinder for M48A2 launcher. The type II cylinder assembly shall be in accordance with Top Assembly, TA13214E0000, and as specified herein.

3.11 Type III, scissoring cylinder for the bridge. The type III cylinder assembly shall be in accordance with Top Assembly, TA13211E7880, and as specified herein.

3.12 Type IV, tongue cylinder.

3.12.1 Style A, for M48A2 launcher. The style A cylinder assembly shall be in accordance with Top Assembly, TA13214E0090, and as specified herein.

3.12.2 Style B, for M48A5/M60A1 launcher. The style B cylinder assembly shall be in accordance with Top Assembly, TA13214E0080, and as specified herein.

3.13 Type V, overhead cylinder.

3.13.1 Style A, for M48A2 launcher. The style A cylinder assembly shall be in accordance with top assembly, TA13214E0010, and as specified herein.

3.13.2 Style B for M48A5/M60A1 launchers. The style B cylinder assembly shall be in accordance with Top Assembly TA13214E0070, and as specified herein.

3.14 Type VI, ejection cylinder for M48A5/M60A1. Type VI cylinder assembly shall be in accordance with Top Assembly, TA13211E3140, and as specified herein.

3.15 Type VII, locking cylinder for M48A2/M48A5/M60A1 launchers. The type VII cylinder assembly shall be in accordance with Top Assembly, TA13211E3130, and as specified herein.

3.16 Type VIII, release cylinder for M48A2 launcher. The type VIII cylinder assembly shall be in accordance with Top Assembly, TA13214E0030, and as specified herein.

3.17 Workmanship.

3.17.1 Welding and Welders.

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3.17.1.1 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings. Welding shall be done by the shielded electric-arc method. Where embrittlement of the joined metal may result, preheating or annealing or both shall be employed. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding whenever practicable.

3.17.1.2 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code;

AWS D1.1, Structural Welding Code, Section 5, Qualification.
Welding Qualifications of the ASME.

Contractors who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welder equipment operators are capable of producing quality welds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserved the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Disassembly inspection. Failure of any test by the preproduction model cylinder assembly or cylinder assembly or cylinder components shall be cause for disassembly in the presence of a Government representative, to the extent necessary to determine the failure. Each disassembled part shall be inspected in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, and tolerances. Parts not complying with such requirements shall be rejected.

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4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Quality Assurance Provisions (QAP) shown on the drawings. Evidence that any part or component does not comply with the applicable drawings shall be cause for rejection of that part or component. The drawings specify the characteristics requiring (QAP) inspection, the sampling plan, and the basis for acceptance and rejection (see 6.6 and 6.7).

4.1.4 Inspection Equipment. Special final inspection equipment to be loaned to the contractor by the Government (see 6.5), is identified in the Quality Assurance Pamphlet listed on the Top Assemblies and 3.7.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction cylinder assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each cylinder assembly shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each cylinder assembly shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. Each cylinder assembly shall be examined as specified herein for the following defects:

101. Any part or component not in accordance with the QAP requirements as shown on the drawings.
102. Any dimension other than those identified by the QAP requirements shown on the drawings not as specified.
103. Materials not as specified and analysis and tests not available or missing.
104. Identification marking not as specified.
105. Treating and painting not as specified.
106. Stenciling not as specified.
107. Components missing or not as specified.

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4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified herein, conduct all tests at an ambient temperature of from plus 40° F to plus 90° F, and oil temperature shall be between plus 40° F to plus 150° F at pump discharge. Use oil conforming to MIL-H-46170. Tests shall be conducted in the order listed herein.

4.5.2.2 Proof pressure. Cycle the cylinder assembly with the piston unrestrained through not less than five complete cycles to demonstrate operation, stroke, and adjustment. Plug the cap-end port and apply a hydraulic pressure of 7,000 psi for a period of 5 minutes at the rod end of each cylinder assembly except for the type VIII cylinder assembly. If the rod moves out of the cylinder assembly during this test, it will be necessary to restrain the rod. Subject the type VIII cylinder assembly to the same test except that the hydraulic pressure shall be reduced to 5,000 psi. The pressure line should be connected to the side port and the cap port left open for one test. For a second test connect the pressure line to the cap end and the side port shall be left open. Evidence of leakage or damage shall constitute failure of this test.

4.5.2.3 Internal leakage.

4.5.2.3.1 Types I, II, III, and IV cylinder assemblies. Connect the type I, II, III, or IV cylinder assemblies as applicable, to a hydraulic system and place the cylinder assembly in a horizontal position with the cap-end port face up. The piston shall be restrained from bearing against the cap end. Insert a graduated, transparent tube of either plastic or glass, and fasten it vertically to the cap-end cylinder port in such a manner as to prevent leakage. One end of the tube shall be open to atmospheric pressure. Apply a 3,800 psi fluid pressure and maintain this pressure in each cylinder assembly, at the rod end. Allow the fluid bypassing the piston to enter and collect in the vertical tube. Measure the rise of fluid level during a period of not less than 3 minutes. Leakage in excess of the rate specified in table I shall constitute failure of this test.

4.5.2.3.2 Type V and VI cylinder assemblies. Test the type V or VI cylinder assemblies as applicable, as specified in 4.5.2.3.1, except that the cap end instead of the rod end of the cylinder assembly shall be pressurized to 3,800 psi. The piston shall be restrained from bearing against the cylinder head. Insert the graduated tube at the rod end of the cylinder assembly. Measure the rise of fluid level during a period of not less than 3 minutes. Leakage in excess of the rate specified in table I shall constitute failure of this test.

4.5.2.3.3 Type VII cylinder assembly. Test the type VII cylinder assembly as specified in 4.5.2.3.1 except that both rod ends of the cylinder assembly shall be tested individually for internal leakage past the restrained piston. Apply a 3,800 psi pressure to one of the rod-end ports, plug the rod-end port, and permit the hydraulic fluid passing the pressurized piston to collect in a tube inserted in the center port. Repeat the test on the other side of the cylinder

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assembly after reversing the pressurized and plugged ports. Leakage in excess of the rate specified in table I shall constitute failure of this test. As an alternate test to the above, the two rod-end ports may be connected to a common line and pressurized to 3,800 psi. If the combined leakage is greater than that specified in table I for one end of the cylinder assembly, both ends of the cylinder assembly shall be tested individually.

4.6 Inspection of packaging.

4.6.1. Preproduction pack inspection.

4.6.1.1 Examination. Examine the preproduction pack for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.1.2. Tests. The preproduction pack for level A shall be tested as follows by the applicable test method.

4.6.1.2.1 Boxed cylinder assemblies weighing less than 200 pounds. Boxed cylinder assemblies weighing less than 200 pounds shall be subjected to the free-fall drop test in accordance with FED-STD-101, Method 5007.

4.6.1.2.2. Boxed cylinder assemblies weighing more than 200 pounds. Boxed cylinder assemblies weighing more than 200 pounds shall be subjected to the pendulum-impact test in accordance with FED-STD-101, Test Method 5012.

4.6.1.2.3 Crated cylinder assemblies weighing more than 1000 pounds. Crated cylinder assemblies weighing more than 1000 pounds shall be subjected to the railroad-car test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A. The car shall strike a string of five empty cars, with draft gear extended and the brakes set, at a speed of not less than 10 miles or more than 11 miles per hour.

4.6.1.3. Basis for rejection. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking, or bracing, or any visual damage to the cylinder assemblies shall constitute failure of the preproduction pack and shall be cause for rejection.

4.6.2. Quality conformance inspection of pack.

4.6.2.1. Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2. Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3. Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

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108. Cylinder assemblies not fully charged with the specified hydraulic fluid for level A or commercial.
109. Piston rod not fully retracted as specified for level A or commercial.
110. Ports not plugged with steel plugs for level A or commercial.
111. Exposed threads of piston rods not protected with steel thread protectors for level A or commercial.
112. Preservative not applied as specified for level A or commercial.
113. Containers not as specified for level A.
114. Cylinder assemblies not packed in the container required as specified by weight for level A.
115. Blocking, bracing, and anchoring not in accordance with the referenced documents for level A.
116. Crated cylinder assemblies not protected with a shroud as specified for level A.
117. Strapping not zinc coated for level A.
118. Marking missing, illegible, incorrect, or incomplete for level A or commercial
119. Cylinder assemblies not palletized as specified, when required.

5. PACKAGING.

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination and testing within the time frame specified (see 6.2) to prove prior to starting production packaging, that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approved by the Government (see 6.8). The preproduction pack may be prepared utilizing either the preproduction cylinder assembly or the production cylinder assembly. When the preproduction model cylinder assembly is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government when requested by the Government to facilitate comparision between the preproduction model cylinder assembly and the production cylinder assembly.

5.2 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Each cylinder assembly shall be filled with the hydraulic fluid specified in 3.4 and the piston rod fully retracted into the cylinder. All ports shall be plugged with steel plugs. Exposed threads of piston rods shall be protected with steel thread protectors. Exposed portions of the piston rod and any other unpainted exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservatives. The preservative shall conform to the applicable specification list in and shall be applied in accordance with MIL-P-116.

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5.2.2 Commercial. Each cylinder assembly shall be preserved as specified for level A except the contractor may use his commercial type preservative compound in place of the type P-1 preservative, providing the preservative will afford protection against corrosion and deterioration from the contractor to the initial description.

5.3 Packing. Packing shall be level A or Commercial as specified (see 6.2)

5.3.1 Level A.

5.3.1.1 Cylinder assemblies weighing less than 1,000 pounds. Each cylinder assembly weighing less than 1,000 pounds shall be packed in a close-fitting box conforming to PPP-B-621, Class 2. Style 2 boxes shall be used for cylinder assemblies weighing up to 500 pounds and style 7 boxes with a hood shall be used for cylinders exceeding 500 pounds. The cylinder assembly shall be blocked, braced, and anchored within the box to prevent movement in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall be zinc coated, and on boxes provided with skids placed adjacent to the skids.

5.3.1.2 Cylinder assemblies weighing more than 1,000 pounds. Each cylinder assembly shall be packed in a crate conforming to MIL-C-52950, Style B, Type V, and protected with a waterproof shroud, in accordance with the appendix to the crate specification. Blocking, bracing, and anchoring shall be in accordance with the applicable crate specification and MIL-STD-1186.

5.3.2 Commercial. Each cylinder assembly shall be individually packed in containers in accordance with MIL-STD-1188.

5.4 Unitization. Type I, VI, VII and VIII cylinder assemblies packed as specified in 5.3.1.1 and 5.3.2 and designated for shipment to production lines shall be palletized when specified (see 6.2). Pallets shall conform to NN-P-71, type and style optional, size 2, group II. The quantity and method of placement on the pallet shall be as recommended by the contractor as determined by the size and configuration of the boxes. The load shall be secured to the pallet with strapping, conforming to QQ-S-781, Class I, Type I or IV, width and thickness as appropriate for weight of the load.

5.5 Marking. Marking for military levels of protection shall be in accordance with MIL-STD-129. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The cylinder assemblies are intended to be used as components of production bridges and production M48A5/M60A1 Bridge Launchers and as replacement/spare parts for M48A2/M48A5/M60A1 Bridge Launchers and bridges currently in the army inventory.

6.2 Ordering data. Acquisition documents should specify the following:

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(a) Title, number, and date of this specification.
(b) Type and style required (see 1.2).
(c) Date of issue of DoDISS applicable to this contract and exceptions thereto (see 2.1.1).
(d) Time frame required for submission of preproduction model (see 3.2).
(e) When Government-loaned property is not required (see 3.7).
(f) Time frame required for submission of the preproduction pack (see 5.1).
(g) Degree of preservation and packaging and degree of packing required (see 5.2 and 5.3).
(h) When palletization is required (see 5.4).

6.3 Preproduction model. Any changes or deviations of production cylinder assemblies from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish cylinder assemblies conforming to this specification.

6.4 Contractors. The contracting officer shall stipulate in the contract that all hydraulic cylinders furnished under terms of any one contract shall be manufactured by the same contractor.

6.5 Government-loaned property. The contracting officer should arrange to loan the inspection equipment specified in 3.7 when applicable.

6.6 Quality assurance provisions (QAP). The contracting officer should require the contractors to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.7 Definition.

6.7.1. Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements Section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.8 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the cylinders in accordance with this specification.

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6.9 Recycled material. It is encouraged that recycled material be used when practical as long as the recycled material meets all requirements of this specification. (see 3.3).

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Review activity:
Army - AT

Preparing activity:
Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-52464B(ME) Cylinder Assemblies, Actuating Linear: Hydraulic, For Bridge and Bridge Launcher

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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